



## Research Paper

The complete degradation of 1,2-dichloroethane in *Escherichia coli* by metabolic engineering

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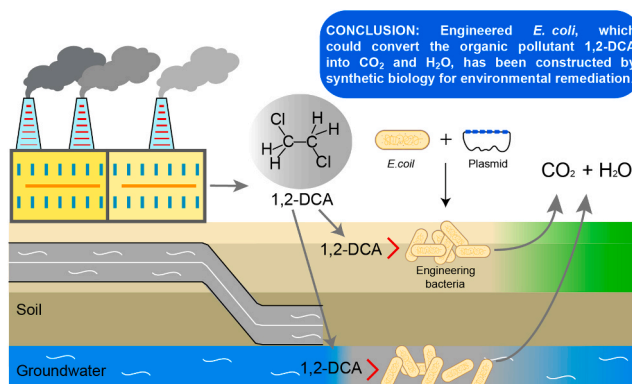
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## HIGHLIGHTS

- The complete 1,2-dichloroethane degradation pathway was assembled into one vector.
- The vector was introduced into the model strain of *Escherichia coli*.
- Engineered bacteria could completely degrade 1,2-dichloroethane.
- The integrity of the degradation pathway was verified by isotope tracing method.

## GRAPHICAL ABSTRACT



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## ABSTRACT

1,2-Dichloroethane (1,2-DCA), a widely utilized chemical intermediate and organic solvent in industry, frequently enters the environment due to accidental leaks and mishandling during application processes. Thus, the in-situ remediation of contaminated sites has become increasingly urgent. However, traditional remediation methods are inefficient and costly, while bioremediation presents a green, efficient, and non-secondary polluting alternative. In this study, an engineered strain capable of completely degrading 1,2-DCA was constructed. We introduced six exogenous genes of the 1,2-DCA degradation pathway into *E. coli* and confirmed their normal transcription and efficient expression in this engineered strain through qRT-PCR and proteomics. The degradation experiments showed that the strain completely degraded 2 mM 1,2-DCA within 12 h. Furthermore, the results of isotope tracing verified that the final degradation product, malic acid, entered the tricarboxylic acid cycle (TCA) of *E. coli* and was ultimately fully metabolized. Also, morphological changes in the engineered strain

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and control strain exposed to 1,2-DCA were observed under SEM, and the results revealed that the engineered strain is more tolerant to 1,2-DCA than the control strain. In conclusion, this study paved a new way for humanity to deal with the increasingly complex environmental challenges.

## 1. Introduction

1,2-Dichloroethane (1,2-DCA) is a synthetic chlorinated aliphatic hydrocarbon (CAHs) extensively utilized in the production of vinyl products and as precursor materials for organic solvents [34,42,50]. A large amount of 1,2-DCA is released into the environment with the industrial boom, especially in soil and groundwater [1,2]. Statistically, the combined annual output of 1,2-DCA in the United States and Western Europe exceeds 17.5 million tons [10]. 1,2-DCA is prone to accumulate in groundwater and form dense non-aqueous phase liquid pools due to its high density and low solubility, leading to environmental pollution [15]. Meanwhile, 1,2-DCA is not easily degraded in nature and will persist for a long time. More importantly, long-term exposure to 1,2-DCA can cause many harmful effects to human, such as central nervous system, liver and lung damage [8,14]. Moreover, the International Agency for Cancer Research (IARC) recognizes 1,2-DCA as a potential human carcinogen [22].

Researchers around the world are actively exploring effective degradation techniques for 1,2-DCA in the environment due to its harmful effects on the environment and health. The current physical and chemical methods have been investigated for reducing the 1,2-DCA from nature include adsorption, extraction, photochemical catalysis and persulfate-based oxidation. Nevertheless, these traditional techniques tend to be costly, labor-intensive, less environmentally friendly, and sometimes inefficient [6,25]. In contrast, the emerging field of bioremediation offers promising alternatives. Bioremediation utilizes microorganisms or plants to break down environmental pollutants, and this approach is sustainable, environmentally friendly, and cost-effective [12,38]. As these microorganisms metabolize, processes like biotransformation and biodegradation degrade organic pollutants in ecosystems. At present, bioremediation has been used to remove a variety of organic pollutants, like nitroaromatic compounds, phenolic substances and heavy metal pollutant [11,7,9]. Moreover, recent studies indicate that directly introducing specific microorganisms into polluted water or soil can also achieve various degrees of remediation [30,33].

Although native microorganisms have strong innate capabilities, the demand for biological scavengers with stronger adaptability, wider substrates, and higher tolerance is constantly increasing [20,31]. However, biological evolution is a long and complex process, and the growth of microorganisms is easily limited by the surroundings environment [39]. Further, the ultimate goal of the native degradation pathway is often survival of microorganisms rather than efficient detoxification, which is unreliable for industrial applications. Therefore, it is crucial to find a way to overcome these difficulties in a relatively short time. Synthetic microbial scavengers created through synthetic biology and metabolic engineering techniques have excellent tolerance and metabolism of pollutant [26,35,47]. Meanwhile, high-throughput sequencing technology and multi-omics analysis tools offer the greater possibilities to design and optimize novel pathways to improve the effectiveness of pollutant degradation [4,48]. Artificially synthetic biological elements have been successfully incorporated into microbial frameworks, producing efficient metabolic pathways to obtain complete mineralization of pollutant. In *E. coli* BL21, metabolic pathways for the degradation of 3-chlorocatechol and 4-chlorocatechol have been successfully reconstructed. These pathways transformed contaminants into compounds like acetyl-CoA and succinyl-CoA, which were then processed through the tricarboxylic acid (TCA) cycle [39]. Using the T7 promoter, 10 foreign enzymes from *p. putida* were expressed in the engineered bacteria. The strain was able to completely degrade 4 mM nitrobenzene in 8 h and 1 mM nitrobenzene in soil within 10 h [9].

In this study, we successfully introduced a complete metabolic pathway for the degradation of 1,2-DCA into *E. coli*, employing a synthetic biology approach for the first time. Six genes of this exogenous pathway were codon optimized and synthesized by chemical methods [46]. The genes *XaDhlA*, *XaADH*, *XaALDH*, and *XaDhlB* from *Xanthobacter autotrophicus* GJ10, the gene *CrGYD1* from *Chlamydomonas reinhardtii*, and the gene *GmMS* from *Glycine max*. Employing the T7 promoter, we modulated the transcription patterns of these genes and integrated them into a vector, thus engineering a bacterium specialized in 1,2-DCA degradation. Our results demonstrate that 1,2-DCA was first degraded to ethanoic acid and then catalyzed to malate subsequently entered the TCA cycle of the bacteria. This study not only constructs strains to degrade 1,2-DCA, but also provides new insights and ideas for creating highly efficient synthetic biological scavengers.

## 2. Materials and methods

### 2.1. Reagents, plasmids, and strains

Unless otherwise specified, all reagents and chemicals employed in this experiment were of analytical grade, sourced from Sangon Biotech (Shanghai, China) Co., Ltd. Total RNA was isolated using the RNA extractor (TRIzol) kit (Sangon Biotech Co., Ltd, Shanghai, China). The TransScript One-Step gDNA removal and first-strand cDNA Synthesis SuperMix came from TIANGEN Biotech (Beijing, China) Co., Ltd. The primers for polymerase chain reaction (PCR) were designed using Primer Premier 5 program and delivered to Sangon Biotech (Shanghai, China) Co., Ltd for synthesis. [ $^{13}\text{C}$ ]-Glycolate used for isotope tracing technique was purchased from Cambridge Isotope Laboratories, Inc. (Massachusetts, USA).

Strain DH5 $\alpha$  (Invitrogen, USA) and pGEM-T easy plasmid (Promega, China) were used for vector construction and molecular clone. For the construction and expression of the synthetic gene cluster, we used the pET-28a-c (+) plasmid (Novagen, USA) in conjunction with the *E. coli* strain BL21-AI (Invitrogen). Our research incorporated six genes sourced from various organisms: *XaDhlA*, *XaADH*, *XaALDH*, and *XaDhlB* were isolated from *Xanthobacter autotrophicus* GJ10, while *CrGYD1* came from *Chlamydomonas reinhardtii* and *GmMS* from *Glycine max*. Nucleotide sequences of all genes were searched and obtained from National Center for Biotechnology Information (NCBI). Furthermore, these genes were optimized to improve their efficient of expression as previously described [40]. The codons of these genes have been optimized based on the preferred codon usage in *E. coli*. Additionally, common restriction enzyme recognition sites within the genes have been removed to facilitate subsequent vector construction. To enhance mRNA stability, reverse repeat sequences, stem-loop structures, and transcription termination signals have been eliminated, and the GC content has been balanced. The details of the optimized nucleotide sequences can be found in the Supplementary Files 1. Using the PCR-based long gene synthesis method, we chemically synthesized the optimized genes, seamlessly connecting them with the T7 promoter and terminator [46]. We utilized classical molecular biology techniques to assemble these vectors. The resultant transformants were labeled BL-5673, while those containing the blank vector were designated BL-5701.

### 2.2. Biodegradation and cultivation conditions

All bacteria in this study were cultured in M9 (pH=7) medium. This medium comprises 0.147 mg calcium chloride dihydrate, 0.5 g sodium chloride, 1.204 g magnesium sulfate heptahydrate, 3 g potassium

dihydrogen orthophosphate, 1 g ammonium chloride, and 12 g disodium hydrogen phosphate heptahydrate per liter. In addition, 10 g L<sup>-1</sup> glycerol, 1 g L<sup>-1</sup> hydrolyzed casein and 33.7 µg L<sup>-1</sup> thiamine hydrochloride (vitamin B1), were added as supplementary nutrients. 2 g L<sup>-1</sup> arabinose and 1 mM isopropyl-β-D-thiogalactoside (IPTG) were used as induction agents, and 30 µg mL<sup>-1</sup> kanamycin sulfate was utilized as screening antibiotics. 100 µg L<sup>-1</sup> pyrroloquinoline quinone (PQQ) and 10 mg L<sup>-1</sup> phenazine methosulfate (PMS) were used in M9 medium during the degradation stage.

After growing at 160 rpm and 37 °C for 24 h, the bacterial cells were collected by centrifugation (4 °C, 6000 rpm, 5 min). The bacterial cells were washed with distillation-distillation H<sub>2</sub>O (ddH<sub>2</sub>O) for three times and re-suspended in M9 medium containing antibiotic and inducer (OD<sub>600</sub> = 2.0). 20 mL of the aforementioned culture were collected to assess its capability to degrade varying concentrations of 1,2-DCA. Additionally, to assess the growth curves of BL-5673 and BL-5701, the harvested cells were re-suspended in M9 medium containing antibiotics, inducers, and 1,2-DCA to an OD<sub>600</sub> of 0.05.

The capacity for degradation by BL-5673 introduced into soil was assessed. The soil was sourced from river mud in Shanghai, China. It was sterilized at 121 °C for 40 min, air-dried, and sieved (0.45 mm). The concentration of 1,2-DCA in the soil was adjusted to 2 mM, and then inoculated with the aforementioned bacterial suspension to achieve a soil to liquid ratio of 1 g/1 mL. The samples were incubated at 30 °C and sampling was conducted every 2 h.

The key metabolite of the degradation process, [U-<sup>13</sup>C]-glycolate (2 mM), was used to replace 1,2-DCA to verify the integrity of the pathway. The ionic fragment with increased molecular weight, including all carbon atoms, was used for the quantitative analysis of isotopologues. The rate of different isotopologues was calculated by analyzing the selected ionic fragment, and the interference of natural rate was eliminated by calculation.

### 2.3. Gene transfer and expression analysis

As described above, *E. coli* underwent genetic transformation and culturing using traditional microbial biotechnology methods. Plasmids from the transformants were then authenticated using the Sanger sequencing technique. RNA was isolated from the engineered bacteria, and cDNA synthesis was conducted as per the provided instructions. The expression levels of genes were analyzed using Quantitative Real-time PCR (qRT-PCR), with the 16 S rRNA serving as the internal control. Relative expression levels of the examined genes were determined according to the 2<sup>-ΔΔCT</sup> method [23]. The primers utilized in this research can be found in Supplement File.

### 2.4. Metabolite identification

Quantification of 1,2-DCA was performed using gas chromatography (GC) with a DB-5 capillary column. Samples were filtered using a 0.45 µm pore size nitrocellulose membrane filter and added to the GC sample vials. The mobile phase was Helium with a flow of 2 mL min<sup>-1</sup>. The column temperature was maintained at 40 °C for 2 min, ramped up to 80 °C at 10 °C min<sup>-1</sup> and maintained for 1 min. The content of 2-chloroethanol was detected using headspace/gas chromatography-mass spectrometry (HS/GC-MS). GC conditions: the DB-624 quartz capillary column was used, with the column temperature maintained at 40 °C for 3 min, then increased by 25 °C min<sup>-1</sup> to 240 °C and held. High-purity He served as the carrier gas with a flow rate of 1.0 mL min<sup>-1</sup> and a split ratio of 20:1. The injection port temperature was 220 °C. MS conditions: ionization energy was 70 eV, ion source temperature was 260 °C, quadrupole temperature was 150 °C, with a solvent delay of 2.0 min. Peak area were used for quantitation.

The concentrations of chloroacetate and glycolate were detected by high-performance liquid chromatography (HPLC) with a C18 column. The detection of chloroacetate was performed at a column temperature

of 30 °C, using a mobile phase of 0.05 % phosphoric acid and methanol (95:5 vol ratio) at a flow rate of 0.8 mL min<sup>-1</sup>, with a detection wavelength of 215 nm. For glycolate detection, the column temperature was set at 35 °C, with a mobile phase of 0.06 % phosphoric acid and acetonitrile (95:5 vol ratio) at a flow rate of 0.6 mL min<sup>-1</sup>, with a detection wavelength of 210 nm. Peak area were used for quantitation.

Organic acids in the tricarboxylic acid (TCA) cycle were analyzed using gas chromatography-mass spectrometry (GC-MS) employing an HP-5MS capillary column. The GC operational settings were as follows: Helium served as the carrier gas at a flow rate of 1.0 mL min<sup>-1</sup>, with an injection volume of 1 µL and a split ratio of 20:1. The column oven temperature began at 80 °C, remained constant for 0 min, then increased to 150 °C at 8 °C min<sup>-1</sup> and subsequently to 280 °C at 10 °C min<sup>-1</sup>, where it was maintained for 20 min. Regarding MS settings, the ion source was bombarded by electrons with an energy setting of 70 eV, and the ion source and quadrupole temperatures were maintained at 230 °C and 150 °C, respectively.

### 2.5. Proteomic assay

Based on the previous description, protein extraction, quantification, and digestion were performed [21]. The peptide mixture was dissolved in mobile phase A (100 % water, 0.1 % formic acid), followed by centrifugation, and the supernatant was collected for liquid chromatography-mass spectrometry (LC-MS/MS) analysis. Mobile phase B (80 % acetonitrile, 0.1 % formic acid) was employed for liquid chromatography elution. Analysis was performed using the ORBITRAP ECLIPSE mass spectrometer (Thermo Fisher Scientific, USA) with a compensation voltage (CV) alternating between -45 and -65 every one second. The ion spray voltage was set at 2 kV, and the ion transfer tube temperature was maintained at 320 °C. Mass spectrometry was conducted using a data-dependent acquisition mode, with a full scan range of 350–1500 m z<sup>-1</sup>. The first-level mass spectrometry had a resolution set to 200 m z<sup>-1</sup>, AGC (automatic gain control) was set to 4 × 10<sup>5</sup>, and the maximum injection time was 50 ms. For the second-level mass spectrometry, the "Top Speed" mode was employed, with a resolution of 200 m z<sup>-1</sup>, AGC set to 5 × 10<sup>4</sup>, and a maximum injection time of 22 ms. The peptide fragmentation collision energy was set to 33 %, generating raw data for the detection of tandem mass spectra.

Data generated by LC-MS/MS were searched using Proteome Discoverer 2.4 (ThermoFisher Scientific, USA) against the Uniprot *E. coli* database (2023/4/18, 4403 sequences) and six exogenous protein sequences. The criteria used for protein identification included a 1 % false discovery rate (FDR) and at least one peptide match for each protein. Protein quantification was conducted based on the peak intensities of unique peptide ions in the LC-MS/MS spectra. The principal component analysis (PCA) was performed on each group of proteomic data using the "FactoMineR" package in the R software [19]. The statistical analysis of differentially expressed proteins was performed using the "limma" package in the R software [29]. The Gene Ontology (GO) annotation for differentially expressed genes was performed using the Gene Ontology database (GO, <http://geneontology.org/>), and the pathway enrichment analysis was conducted using the Kyoto Encyclopedia of Genes and Genomes database (KEGG, <https://www.genome.jp/kegg/>). Protein associations and inferred interactions (kappa > 0.7) were collected from the STRING database (<https://string-db.org/>). The protein-protein interaction (PPI) network graph was constructed using Cytoscape (v3.10), with nodes representing the name of proteins, and the nodes size indicates their significance level.

### 2.6. Data analysis and visualization

All data are expressed as mean ± standard deviation (SD) and shown by error bars. One-way ANOVA (*p* < 0.05) and Tukey's HSD test (*p* < 0.05) were performed using the R software package [36]. The "ggplot2" package was used to visualize statistical data [43].

### 3. Result

#### 3.1. Optimization and assembly of 1,2-DCA degradation pathway

Nucleotide sequences of six genes involved in 1,2-DCA metabolism were optimized for the assembly of the complete degradation pathway. The optimized genes were named as *XaDhlAS*, *XaADHS*, *XaALDHS*, *XaDhlBS*, *CrGYD1S* and *GmMSS*, and the homology of the optimized nucleotide sequences with the original sequences were 79.57 %, 80.49 %, 82.68 %, 81.27 %, 80.23 % and 81.08 %, respectively. All genes were synthesized by chemical synthesis and spliced to T7 promoters and terminator seamlessly. Each gene was an independent expression unit, and all these units were assembled into a metabolic module (Fig. 1). Gel electrophoresis and qRT-PCR analysis confirmed the successful transcription of all genes within the module in the engineered bacteria (Fig. 2).

#### 3.2. Degradation of 1,2-DCA by the transformant BL-5673

To assess the degradation capacity of the engineered strain BL-5673, it was exposed to various concentrations of 1,2-DCA. This strain effectively degraded 1 mM of 1,2-DCA within 8 h and 2 mM within 12 h. In comparison, the 1,2-DCA levels in the control group remained relatively unchanged over the same duration (Fig. 3A). Additionally, the concentration of the intermediate metabolites 2-chloroethanol, chloroacetate, and glycolate were detected. The results indicated that the concentrations of all three metabolites exhibited an initial increase followed by a decrease. Notably, a significant accumulation of glycolate at 4 h was observed, likely due to its synthesis rate surpassing its metabolic rate (Fig. 3B, Supplement File). Furthermore, with the increase in the concentration of 1,2-DCA, the degradation efficiency of the engineered bacteria gradually decreases. When the concentration of 1,2-DCA in the culture medium is 4 mM, the engineered bacteria can degrade 72.4 % of the 1,2-DCA, whereas at a concentration of 6 mM, the engineered bacteria can only degrade 43.6 % of the 1,2-DCA (Fig. 3C). This phenomenon could potentially be attributed to the inhibitory effect of elevated concentrations of 1,2-DCA on the normal physiological processes of the bacterial life cycle. Environmental factors can influence the microbial capacity for pollutant degradation [27]. Therefore, we introduced 1, 2-DCA into the soil and inoculated it with BL-5673 to simulate the pollutant degradation capability of engineered bacteria under natural conditions. The results suggest that, despite a slight deceleration in the degradation efficiency of engineered bacteria in soil, they were still

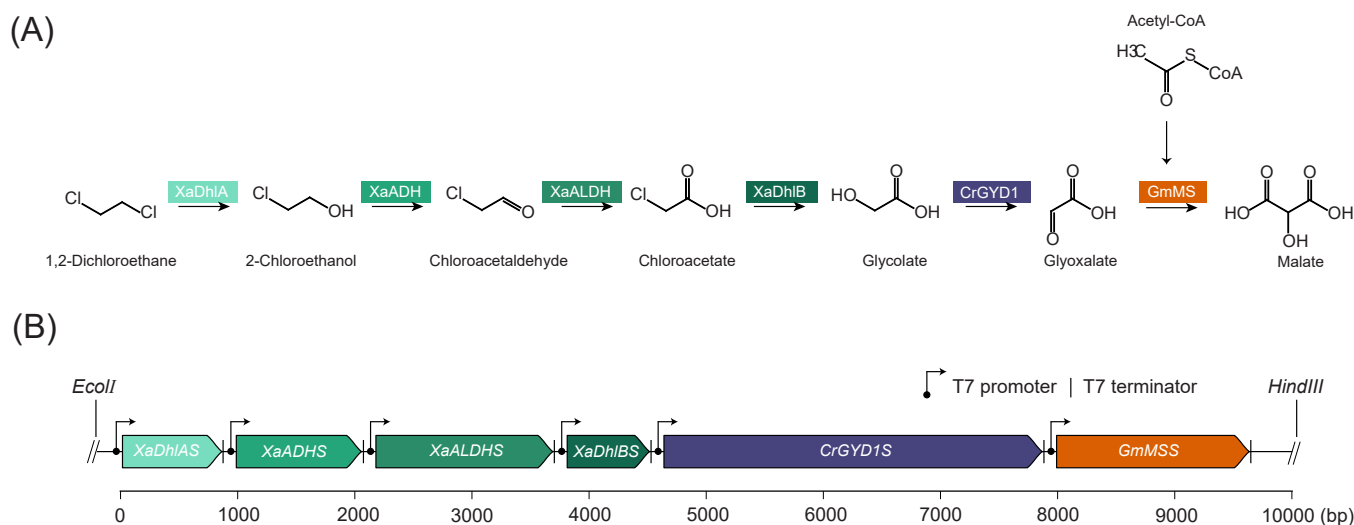
capable of completely degrading 1,2-DCA within 16 h (Fig. 3D).

The above results have shown changes in the concentration of glycolate, demonstrating that 1,2-DCA was degraded to glycolate according to the constructed metabolic pathway. We further verified that glycolate was completely degraded using isotopic tracing. Using  $[U-^{13}C]$ -glycolate as substrate, the  $^{13}C$  abundance of organic acids in the TCA cycle were measured by GC-MS, proving that the final degradation product, malate, entered into the TCA cycle of *E. coli* itself smoothly. The final metabolites of TCA were carbon dioxide and water, verifying that the complete degradation of 1,2-DCA (Fig. 4).

#### 3.3. Growth and physiological responses of BL-5673 exposure to 1,2-DCA

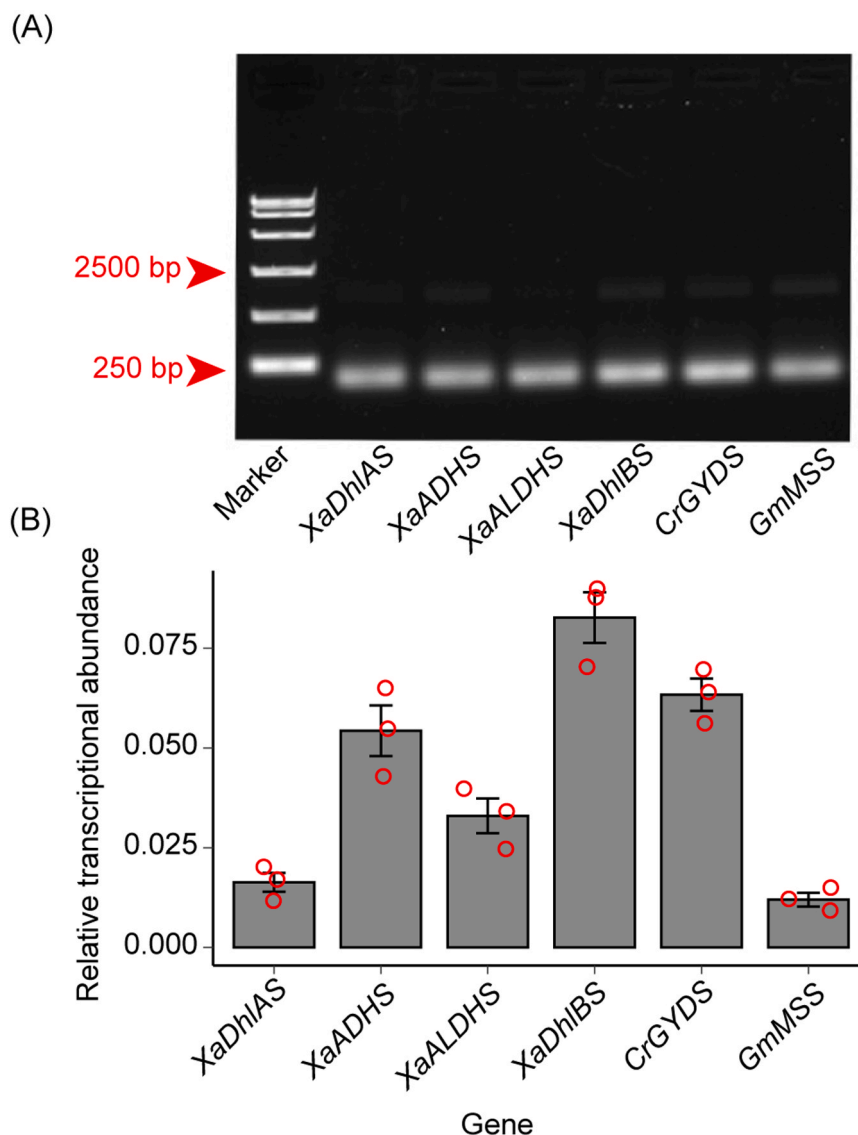
To investigate the impact of exogenous genes on protein expression in *E. coli*, total proteins were extracted and quantified from BL-5701, BL-5673 and BL-5673+ 1,2-DCA (representing BL-5673 treated with 1,2-DCA) after 12 h of growth using label-free quantitative proteomic strategy. A total of 2121 proteins were identified. Principal Component Analysis (PCA) distinguished BL-5701 from both BL-5673 and BL-5673+ 1,2-DCA. However, the distinction between BL-5673 and BL-5673+ 1,2-DCA was minimal (Fig. 5A). Additionally, the exogenous proteins were effectively expressed in engineered bacteria, demonstrating a high level of protein abundance (Fig. 5C). Using a threshold of  $p$ -value < 0.05 and fold-changes > 1.2, a total of 419 differentially expressed proteins (excluding exogenous genes) were identified between BL-5701 and BL-5673. Compared to the BL-5701 strain, 223 proteins were upregulated and 196 proteins were downregulated in BL-5673. However, only 29 differentially expressed proteins were observed between BL-5673 and BL-5673+ 1,2-DCA. (Fig. 5C-D). Furthermore, there was a significant correlation in the total protein abundance between BL-5673 and BL-5673+ 1,2-DCA (Pearson's correlation coefficient ( $R$ ) = 0.99) (Fig. 5E).

In order to deeply investigate the effects of exogenous genes on the physiological activities of *E. coli* itself, enrichment analysis was performed on the differentially expressed proteins between BL-5701 and BL-5673. The Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis indicated that the differentially expressed proteins were predominantly enriched in the fundamental metabolic processes that maintain bacterial life activities, including ABC transporters, pyruvate metabolism, ascorbate and aldarate metabolism (Fig. 5F). In addition, protein-protein interaction (PPI) networks for the upregulated and downregulated genes were constructed using the String database (Fig. 5G). The network graph of upregulated genes is composed of 186



**Fig. 1.** 1,2-Dichloroethane (1,2-DCA) degradation construction. (A) Complete 1,2-DCA degradation pathway constructed in this study. (B) Schematic of recombinant vectors used in this study.





**Fig. 2.** Expression verification of engineered genes. (A) Polymerase chain reaction (PCR) identification of all genes on the expression vector. The PCR reaction was carried out using plasmid extracted from BL-5673 as template. (B) RNA-level expression of all synthetic genes in transformants BL-5673. Fluorescence quantitative PCR was performed using the cDNA of BL-5673 as template.

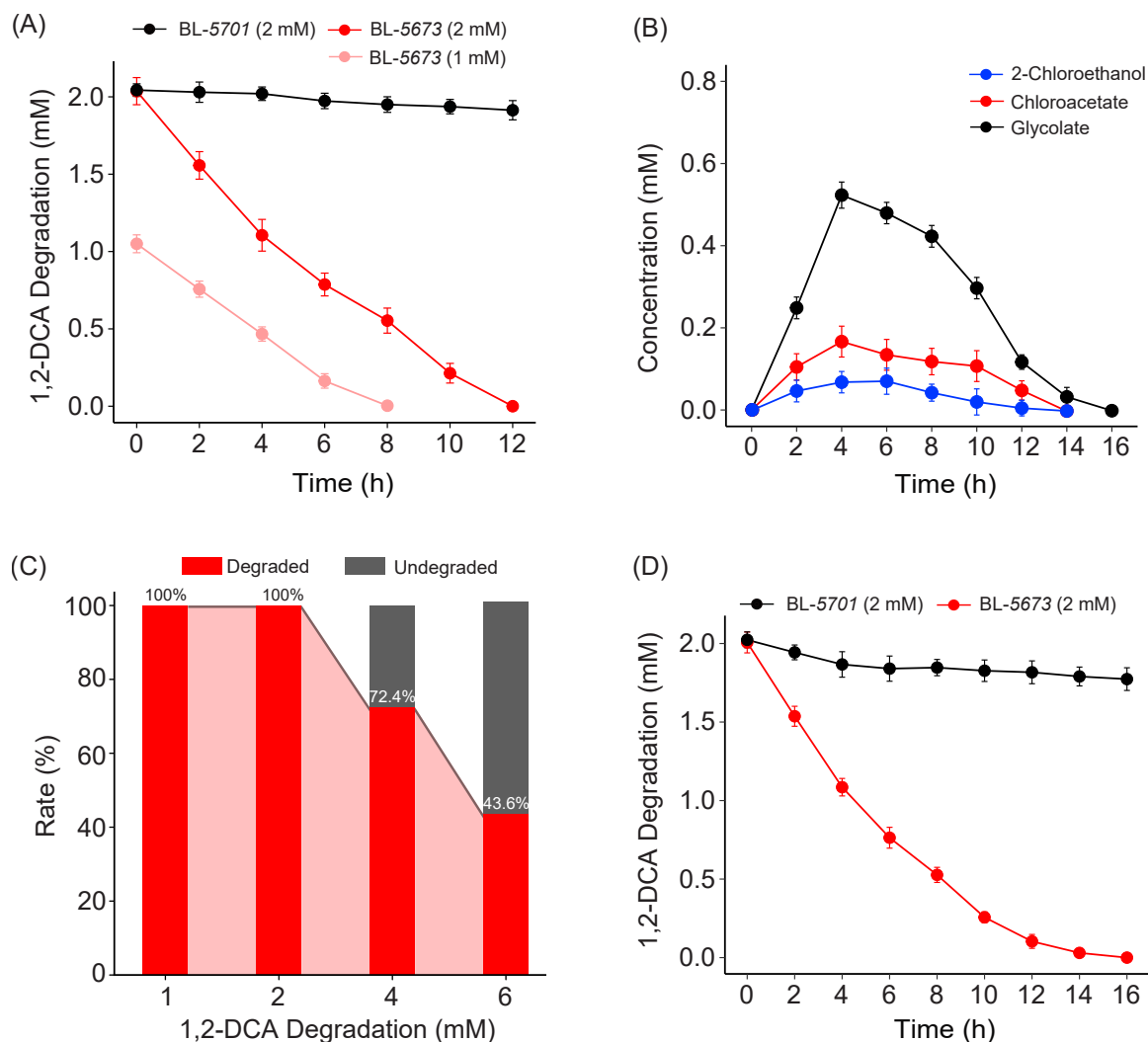
hypothetical edges, whereas the network graph of downregulated genes consists of 233 hypothetical edges. Gene Ontology (GO) enrichment analysis classified the two PPI networks into 7 protein clusters. The upregulated genes were enriched in biological processes of monocarboxylic acid metabolic, carboxylic acid catabolic, spermidine biosynthetic, methionine biosynthetic and methylation. Among them, spermidine is a crucial substance for regulating the formation of microbial biofilms, and the formation of biofilms can enhance microbial resistance to external stimuli [13,17]. Methionine is an essential sulfur-containing amino acid that plays an indispensable role in protein synthesis, catalytic function regulation, and post-translational modifications [24]. Furthermore, the downregulated genes were enriched in biological processes of protein fold and responding to stress. An interesting observation is the negligible enrichment of differential protein expression between BL-5673 and BL-5673+1,2-DCA, indicating that the engineered bacteria have strong tolerance and detoxification ability to 1,2-DCA.

Although the introduction of exogenous genes leads to changes in protein levels, its impact on the growth and reproduction of *E. coli* remains unknown. Dynamically monitored the growth curves of BL-5673

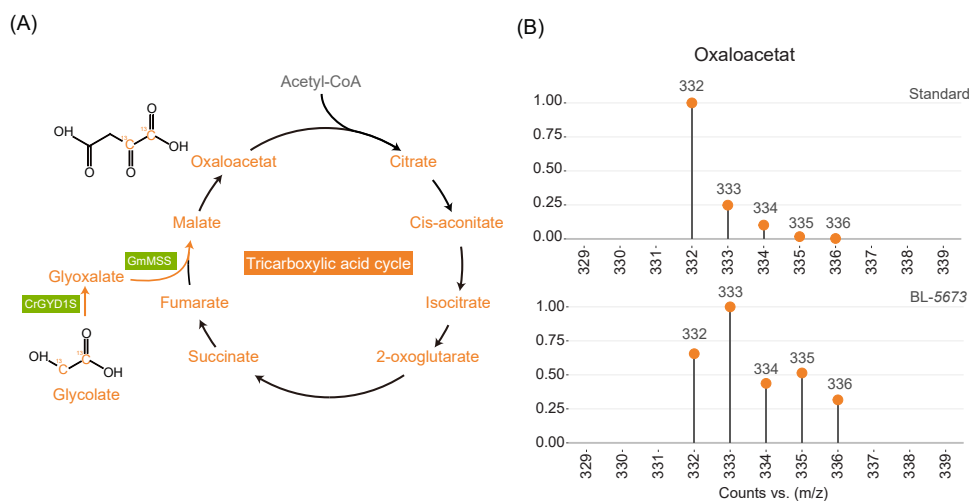
and BL-5701 exposed to 1 mM 1,2-DCA and under normal growth conditions (Fig. 5 H). It is worth noting that the introduction of exogenous genes did not exert a significant impact on the growth of the engineered bacteria. After 48 h of cultivation, OD<sub>600</sub> of BL-5701 and BL-5673 reached 1.30 and 1.26, respectively. Although BL-5673+1,2-DCA exhibited a minor growth delay in the early stages, there was no significant difference in growth status between BL-5673+1,2-DCA and BL-5673 at 48 h. This phenomenon further corroborates the findings of proteomics, confirming the growth stability of engineered bacteria in contaminated media. In contrast, the growth of BL-5701 exposed to 1,2-DCA was significantly inhibited, with the OD<sub>600</sub> at 48 h being only 0.57-fold compared to the control group. These results indicated that the engineered bacteria possess excellent resistance and detoxification capabilities to 1,2-DCA.

### 3.4. Morphological observation of bacteria

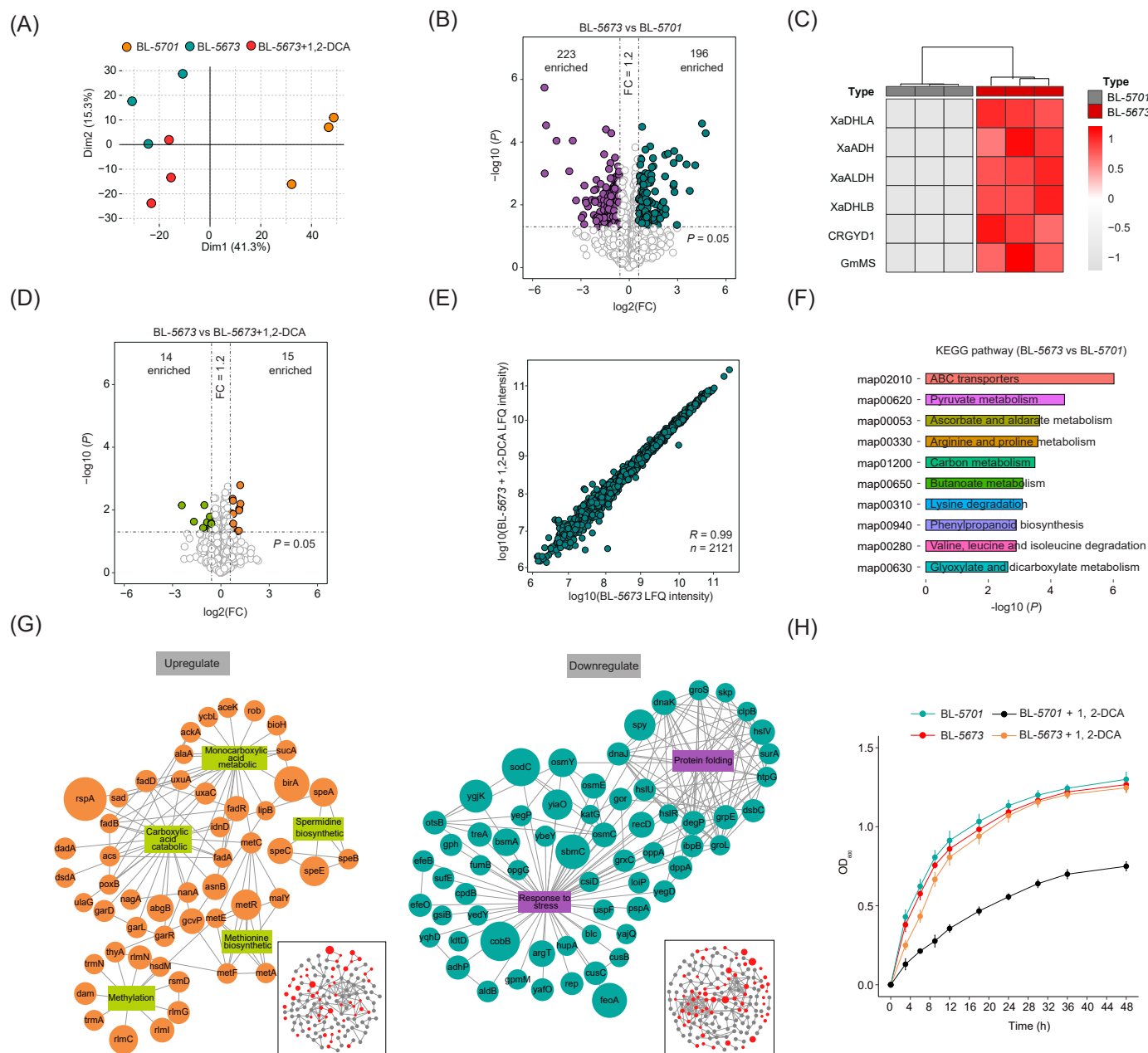
We further observed the morphological changes of BL-5701 and BL-5673 when exposed to 1,2-DCA. After 6 h of exposure, some cells of both strains became misshapen ellipses with unclear outlines. This could be



**Fig. 3.** Degradation of 1,2-DCA by engineered bacterial strains. (A) Degradation of 1,2-DCA by transformants BL-5673 and BL-5701 in nutrient solution. (B) Detection of intermediate products during the degradation of 1,2-DCA. (C) Degradation rate of BL-5673 at different 1,2-DCA concentrations within 12 h. (D) Degradation of 1,2-DCA by transformants BL-5673 and BL-5701 in soil. Error bars denote standard deviation (SD).



**Fig. 4.** Tracing of carbon atoms in glycolate. (A) Schematic diagram of TCA cycle with carbon-13 labeling. (B) Abundance of oxaloacetate <sup>13</sup>C in BL-5673.

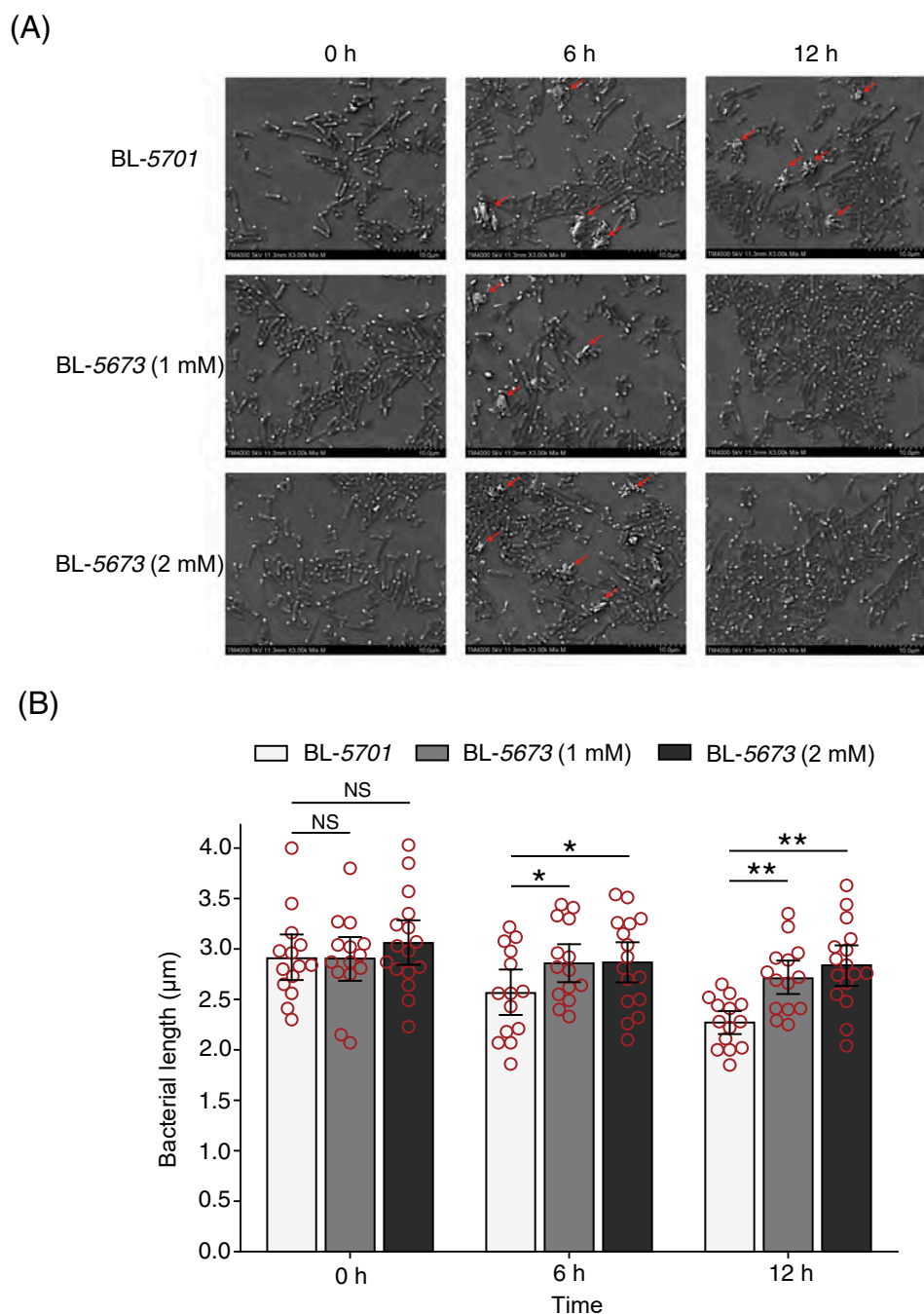


**Fig. 5.** Proteomic analysis and growth condition of engineered bacterial strains. (A) Principal components analysis (PCA) of all proteins detected by mass spectrometry. (B) label free quantitation (LFQ) comparison of proteins detected by BL-5673 and BL-5701. Identification of differentially expressed proteins using thresholds  $P < 0.05$  and fold change (FC)  $> 1.2$ . (C) Expression of exogenous genes in engineer bacteria. (D) As in (B) but for BL-5673 and BL-5673 with 1 mM 1,2-DCA. (E) LFQ correlation analysis of proteins detected by BL-5673 and BL-5673 with 1 mM 1,2-DCA. The symbol  $n$  denotes the total number of proteins detected, and  $R$  represents the Pearson correlation coefficient. (F) Bars illustrate the top ten Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways enriched in differentially expressed proteins between BL-5673 and BL-5701. (G) STRING analysis map of differentially expression protein between BL-5673 and BL-5701. Left, the STRING analysis map of upregulated genes. Right, as in left graph but for downregulated genes. The thumbnail located in the bottom right corner depicts a protein-protein interaction map, where red dots represent enriched proteins. The larger graph represents the protein-protein interaction (PPI) network of enriched proteins and the gene ontology (GO) term for biological process. The node size denotes the significance level of differentially expressed protein (BL-5673 vs BL-5701). (H) The growth condition of BL-5673 and BL-5701 with and without 1 mM 1,2-DCA. Error bars denote standard deviation (SD).

due to the toxicity of 1,2-DCA causing cell damage. Interestingly, after 12 h of exposure, the cells of BL-5673 reverted back to their normal state, while some cells of BL-5701 remained damaged (Fig. 6A). Furthermore, both at 6 and 12 h post exposure, the cell length of BL-5701 was significantly shorter than that of BL-5673 (Fig. 6B). This indicated that, despite the initial damage caused by 1,2-DCA, BL-5673 exhibited a remarkable ability to revert its cell morphology, possibly by neutralizing the toxic effects.

#### 4. Discussion

The development of genetic engineering and synthetic biology has promoted the artificial modification of microorganisms, endowing them with special abilities. By artificially modifying at the genetic and protein levels, the target metabolic pathway can be more efficient. This ultimately makes these engineered microorganisms cost-effective, efficient and widely applicable to bioremediation [44]. Successful cases of using this technology to remediate polluted environments have been reported



**Fig. 6.** Morphological changes of bacteria exposed to 1,2-DCA. (A) The Scanning Electron Microscopy (SEM) was used to observe BL-5701 and BL-5673 after exposure to 1 mM 1,2-DCA and 2 mM 1,2-DCA, respectively. (B) Measurement of *E. coli* cell length after exposure to 1,2-DCA. Error bars denote standard deviation (SD). \* and \*\* respectively indicate least significant differences at the 0.05 and 0.01 probability level compared with BL-5701. NS indicates not significant ( $P > 0.05$ ).

[9,52]. In this study, we have successfully constructed a new engineered *E. coli* strain that can efficiently and greenly degrade 1,2-DCA.

Enzymes and pathways with biological potential can be transplanted from their native, non-engineerable hosts into genetically tractable chassis organisms by combining genetic engineering techniques. However, these genes are often not closely linked and may be distributed on different plasmids or even different organisms [41]. For example, the genes responsible for the complete degradation of phenol in engineered *E. coli* originate from four different microorganisms [40]. By serially combining genes from *Ralstonia* JMP134, *Rhodococcus* sp. DK17, and *Rhodococcus* sp. PBTS2 strains, an engineered bacterium capable of complete degradation of 2,4-dichlorophenoxyacetic acid was

constructed [41]. The four genes (*XaDh1A*, *XaADH*, *XaALDH*, *XaDh1B*) used in this study, involved in the degradation of 1,2-DCA, are all derived from *Xanthobacter autotrophicus* GJ10, and they are capable of converting 1,2-DCA into glycolate. The other two genes, *GrGYD*, *GmMS*, are derived from *Chlamydomonas reinhardtii* and *Glycine max*, respectively, and are able to convert glycolate into malate. Overexpressing genes via plasmid is a primary method for constructing degradative engineered bacteria. However, introducing an excessive number of plasmids into the same strain can impose a heavy burden on the cells and reduce the growth and reproduction rate of the strain [5,44]. Assembling genes from different organisms into a single plasmid and introducing them into bacteria for modification is considered a viable



approach. Each concatenated gene needs to undergo chemical synthesis and codon optimization to better adapt to the new host organism. In this study, we constructed a complete pathway for the degradation of 1, 2-DCA, and each gene was seamlessly linked with the T7 promoter and organized in the form of gene clusters. The degradation rate of 1, 2-DCA is constrained by the chloroacetaldehyde and chloroacetic acid metabolism rates, and these intermediates are more toxic and less available compared to 1,2-DCA. Therefore, efficient and complete degradation of 1,2-DCA is necessary. In this study, the BL-5673 strain could completely degrade 1 mM of 1,2-DCA within 8 h. The intermediate product, glycolate, showed an accumulation followed by a decrease trend. Furthermore, the isotope tracing method demonstrated the integrity of this pathway, and malate can enter the TCA cycle for utilization by the bacteria.

"Foreign" proteins may not be recognized or regulated by cellular components, leading to unforeseen outcomes [49]. In this study, we thoroughly investigated the intrinsic impact of heterologous genes on *E. coli* through proteomic analysis. Interestingly, the heterologous genes directly enhanced the stability of the engineered bacteria in the presence of pollutants. This could be attributed to the higher efficiency of the degradation pathway, which reduced the toxic effects of pollutants on the engineered bacteria. Additionally, a significant number of differentially expressed proteins were observed between BL-5673 and BL-5701, but they did not have a detrimental effect on bacterial growth.

Despite the confirmed ability of engineered bacteria to efficiently degrade 1,2-DCA, there are still some challenges in practical applications. Most of the available data are generated on a laboratory scale, and external factors such as temperature, pH, and nutrient composition can influence the effectiveness of bacterial degradation of pollutants [45]. For example, the degradation pathway of 1,2-DCA is highly aerobic, and lack of oxygen in the subsurface space may limit its normal metabolism. So artificial oxygen supplementation may be necessary to support the normal activity of microorganisms, such as air injection or liquid oxygen injection [18]. In the future, the continued exploration of novel chassis organisms and optimization of metabolic pathways are crucial measures to advance the application of biodegradation technology in environmental pollution control and sustainable development. Fortunately, with the advancement of high-throughput multi-omics phenotypic analysis techniques, machine learning (mL) can offer phenotype predictions even without a comprehensive understanding of the underlying mechanisms [28]. This not only provides design insights for early-stage metabolic pathways but also significantly enhances design efficiency. Furthermore, plasmids are considered to be mobile genetic elements, which inevitably leads to ecological risks due to their potential for non-targeted horizontal gene transfer [3]. For instance, soil microbial strains are capable of degrading 1,2-DCA through plasmid-mediated horizontal transfer in the environment [37]. Soil bacteria have acquired the ability to degrade chlorpyrifos through the acquisition of exogenous plasmids [51]. Various methods have been developed to prevent such occurrences, such as introducing controllable programmed cell death circuits within bacteria or creating engineered bacteria that do not carry antibiotic resistance genes [16,32]. Nevertheless, we must remain vigilant about the actual performance of these strategies in complex environments.

## 5. Conclusion

In this study, synthetic biology technology was utilized to construct engineered strain capable of completely degrading 1,2-DCA. Using the model strain *E. coli* as a chassis cell, an integrated degradation pathway was introduced to realize the complete degradation of 1,2-DCA within 12 h. In the future work, strains with wider substrate range and better adaptability can be selected as chassis cells, and the constructed engineering bacteria can adapt to a wide variety of polluted environments, and ultimately realize full-scenario degradation in different polluted environments. Therefore, the application of synthetic biology

technology to create functional strains that completely degrade 1,2-DCA is very promising and has good economic and ecological benefits.

## Environmental Implication

1,2-Dichloroethane (1,2-DCA), a commonly utilized industrial organic solvent, leads to significant environmental pollution due to its extensive release during large-scale industrial applications. Extensive toxicological research shows that prolonged exposure to 1,2-DCA poses serious health risks to animals, plants, and humans, including carcinogenicity. In this study, an engineered strain that can fully degrade 1,2-DCA was constructed by synthetic biological methods, ultimately achieving in situ bioremediation of 1,2-DCA contaminated areas.

## CRediT authorship contribution statement

**Zhen-Jun Li:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Xiao-Yan Fu:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Jian-Jie Gao:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Bo Wang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Jing Xu:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Zhi-Hao Zuo:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Quan-Hong Yao:** Writing – review & editing, Project administration, Methodology, Formal analysis, Conceptualization. **Hao Zhang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Yong-Dong Deng:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Yong-Sheng Tian:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Wen-Hui Zhang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Ri-He Peng:** Writing – review & editing, Project administration, Methodology, Formal analysis, Conceptualization. **Li-Juan Wang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Yu Wang:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2024.134476](https://doi.org/10.1016/j.jhazmat.2024.134476).



- [50] Zeng, N., Zhang, Z., Jiang, H., Li, R., Chang, C., Wang, F., et al., 2019. LncRNA-241 inhibits 1,2-Dichloroethane-induced hepatic apoptosis. *Toxicol Vitro* 61, 104650. <https://doi.org/10.1016/j.tiv.2019.104650>.
- [51] Zhang, Q., Wang, B., Cao, Z., Yu, Y., 2012. Plasmid-mediated bioaugmentation for the degradation of chlorpyrifos in soil. *J Hazard Mater* 221-222, 178-184. <https://doi.org/10.1016/j.jhazmat.2012.04.024>.
- [52] Zhang, W.H., Deng, Y.D., Chen, Z.F., Zuo, Z.H., Tian, Y.S., Xu, J., et al., 2023. Metabolic engineering of *Escherichia coli* for 2,4-dinitrotoluene degradation. *Ecotoxicol Environ Saf* 262, 115287. <https://doi.org/10.1016/j.ecoenv.2023.115287>.